PANJAB UNIVERSITY, CHANDIGARH

OUTLINES OF TESTS, SYLLABI AND COURSES OF READING FOR M.PHIL IN ECONOMICS (SEMESTER SYSTEM) FOR THE SESSION – 2021-22

Total Credits for M.Phil Course Work over Semester I & II = 28

SEMESTER – I **Total Credit (10)**

Paper-I	Research Publication and Ethics	100 marks (2 credits) (50+50 Internal Assessment)
Paper-II	Research Methodology	100 marks (4 credits) (50+50 Internal Assessment)
Paper-III	Optional any one of the following i) Applied Econometrics ii) Economic Theory iii) Economic Development	100 marks (4 credits) (50+50 Internal Assessment)

SEMESTER-II **Total Credit (8)**

Paper- IV & V: Optional (Any two of the following):

(i)	Public Policy and Governance	100 marks (4 credits) (50+50 internal assessment)
(ii)	Economic Methodology	100 marks (4 credits) (50+50 internal assessment)
(iii)	Industrial Economics	100 marks (4 credits) (50+50 internal assessment)
(iv)	Money, Banking and Finance	100 marks (4 credits) (50+50 internal assessment)
(v)	International Economics	100 marks (4 credits) (50+50 internal assessment)
(vi)	Agricultural Economics	100 marks (4 credits) (50+50 Internal Assessment)
Paper- VI	Dissertation	200 marks (8 credits) (Dissertation 150 + viva-voce 50 marks)

The students will carry out their research work from the beginning of their 1st semester and complete the same within a period of two semesters.

....

SEMESTER-I

PAPER-I: RESEARCH PUBLICATION AND ETHICS

2 Credits
100 marks
(50+50 internal assessment)
Teaching Hour : 40
Time : 3 Hours

Objectives:
This course aims to develop an understanding of ethical issues related to research and publication. The students will acquire skills to write research papers/thesis following publication ethics. Workshops have been designed to internalize ethical issues through in depth discussion in order to identify research misconduct and predatory publications. Indexing and citation databases, open access publications, research metrics (citations, h-index, Impact Factor, etc.) and plagiarism tools will be introduced in this course.

Pedagogy: Class Room Teaching, Term Paper/Assignment, Presentations and Participation in Workshops

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The 1st question consisting of 2 short notes, each carrying 5 marks, set out of the entire syllabus would be compulsory. Additionally, the students are required to attempt one question from each Unit carrying 10 marks.

Unit-I

Ethics and Scientific Conduct

Introduction to Ethics: philosophy, concept, nature and scope

Ethics with respect to science and research. Intellectual honesty and research integrity.

Scientific misconducts: Falsification, Fabrication, and Plagiarism (FFP); salamislicing;

Selective reporting and misrepresentation of data.

Unit-II

Academic Writing

Importance of Good Academic Writing.

Elements of Competent Writing: Purpose, Clarity, Coherence, Argument Building.

Types of Academic Writing; difference between Academic Writing and other Writings. Referencing Style and Bibliography.

Unit-III

Publication Ethics and Misconduct

Publication ethics: definition and importance.

Publication misconduct: definition, concept, problems that lead to unethical behavior and vice versa.

Violation of publication ethics: Redundant publications (duplicate and overlapping publications); Conflict to finterest; Predatory publishers and journals.

Identification of publication misconduct, complaints and appeals.

Standard setting initiatives and guidelines: COPE, WAME, etc.

Unit-IV

Online Publication Tools

Open access publications and initiatives like Sherpa Romeo (online resource to check publisher copyright & self-archiving policies).

Journal finder/ journal suggestion tools viz. JANE, Elsevier Journal Finder, Springer Journal Suggester, etc.

Issues of Copyright: Use of plagiarism software like Turnitin, Urkund and other open source software tools.

Workshops

1) Subject specific ethical issues, FFP, authorship
2) Conflict of interest
3) Complaints and appeals: examples and fraud from India and abroad
4) Indexing databases
5) Citation databases: Web of Science, Scopus, etc.
6) Impact Factor of journal as per Journal Citation Report,SNIP,SJR,IPP,Cite Score Metrics: h-index ,gindex, ilO index, altmetrics
7) Best Practices

References:

1. Bird,A.(2006).*Philosophy of Science.* Routledge.
2. Israel,M. and I.Hay (2006). *Research Ethics in Social Sciences.* Sage
3. Macintyre, Alasdair (1967) *A Short History of Ethics.* London.
4. P.Chaddah, (2018) Ethics in Competitive Research: Donotgets cooped; donot get plagiarized, ISBN: 978-9387480865
5. National Academy of Sciences, National Academy of Engineering and Institute of Medicine. (2009).*On Being a Scientist: A Guide to Responsible Conduct in Research: Third Edition.* National Academies Press.
6. Resnik, D.B.(2011).What Is ethics in research & why is it important. *National Institute of Environmental Health Sciences,* 1-10. Retrieved fromhttps://www.niehs.nih.gov/research/resources/bioethics/whatis/index.cfrn
7. Beall,J.(2012). Predatory publishers are corrupting open access. Nature, 489 (7415), 179-179.
8. Indian National Science Academy (INSA), Ethics in Science Education, Research and Governance (2019), ISBN:978-8l-939482-l-7.https//www. insaindia.res.in/pdf /Ethics

PAPER-II: RESEARCH METHODOLOGY

4 Credits
100 marks
(50+50 internal assessment)
Teaching Hour: 40
Times: 3 Hours

Objectives: The objectives of this paper are (i) to acquaint the students with the advances in the filed of Research Methodology; and (ii) develop research methodological skills for undertaking research in social sciences in general and in a discipline in particular.

Pedagogy: Class Room Teaching, Term Paper/Assignment, Presentations, Book Reviews & Seminars.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The 1st question consisting of 2 short notes, each carrying 5 marks, set out of the entire syllabus would be compulsory. Additionally, the students are required to attempt one question from each Unit carrying 10 marks.

Unit-I

Philosophy of Social Sciences:
Approaches to Social Sciences Research: Quantitative and Qualitative. Inter-disciplinary Research in Social Sciences: Rationale and Approaches.Social Science Research Process.
Selection and Formulation of Research Problem under Quantitative and Qualitative research approach.
Research Synopsis: Concept, Format and Writing.

Unit-II

Hypothesis: Concepts, Types, Features and Testing. Sampling: Concepts, Types, Size, Sampling Error. Triangulation: Concept, Rationale and Process.
Meta Analysis: Concept, Rationale and Process.

Unit-III

Methods of Data Collection.Data Processing
Data Analysis
Report and Thesis Writing.

Unit-IV

Data Sources of Indian Economy
Nature and Sources of Micro and Macro statistical data relating to: Agriculture, Industry, Public Finance, Banking & Finance, External Trade, Consumption, Employment, andNational Income.

Suggested Readings:-

1. Creswell,J.W. (1994). *Research Design: Qualitative and Quantitative Approaches,* London, Sage Publications.
2. Miah, Abdul Quader. (2016). *Applied Statistics for Social and Management Sciences*, (1st ed.) Springer, Singapore.
3. Moser. C.A. and Kalton G. (1979). *Survey Methods in Social Investigation*, (2nd ed.) The English Language Book Society and Heinemann Educational Books. London.
4. Delbert C. Miller. *Handbook of Research Design and Social Measurement*, (5th ed.) Sage Publications, London.
5. David Nachmias and Chara Nachmias. *Research Methods in the Social Sciences*. (2nd ed.) St. Martin's Press Inc, New York.
6. Clarie S.L. *Research Methods in Social Relations*, Holt Rinehart and Winston. NewYork.
7. Pauline V. Young. (2007) *Scientific Social Research and Surveys, India*. (4th ed.) Asia Publishing House.
8. William J. Goode and Paul Hatt. (2006) *Methods in Social Research*, (Ist IndianReprint). USA, Mc Graw K. Hill Book Company.
9. Norman K.Denzin, Yvonna Lincoln. (2003). *The Landscape of Qualitative Research: Theories and Issues*. Sage Publications, New Delhi.
10. Clive Seale, Giampietro Gobo, Jaber I. Gubrium& David Silverman.(2004) *Qualitative Research Practice*. Sage Publications, New Delhi.
11. Dawn Durton. (2000). *Research Training for Social Scientists*, Sage Publications, New Delhi.
12. Royle A.Singleton, Jr. and Bruce C. Straits. (1999). *Approaches to Social ScienceResearch,* Oxford University Press.
13. Mason, Jennifer. (2002). *Qualitative Researching*. Sage.
14. Chandrasekhar C.P.& Jandhyala B.G. Tilak (2001) *India's Social-Economic Database – surveys of selected areas, , Tulika,* New Delhi.
15. Alan Bryman.(2015). *Social Research Methods*. (5thed.) Oxford University Press, United Kingdom.

PAPER III: OPTION (i) – APPLIED ECONOMETRICS

4 Credits
100 marks
(50+50 internal assessment)
Teaching Hour: 40
Time: 3 Hours

Objectives: Application of economic theory needs a reasonable understanding of economic relationships and relevant statistical methods. The econometric theory thus becomes a very powerful tool for understanding of applied economic relationships and for meaningful research in economics. This paper accordingly is devoted to equip the students with basic theory of econometrics and relevant applications of the methods. The topics covered in the course include methods to develop and implement various econometric models (both single equations and simultaneous equation models) in advance statistical/econometrics packages (i.e., STATA/Eviews/SPSS/Gretl/R).

Pedagogy: 50 marks allotted for class room teaching and 50 marks for Lab Practical. The Lab practical comprising 25 marks for successful execution of computer task, 10 marks for computer file, and 15 marks for viva-voce on implementing the econometric techniques in a given package on research topic assigned by the examiner. **Student may use any software among STATA/Eviews/SPSS/Gretl/R to answer the question during written exam and viva- voce**.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The 1st question consisting of 2 short notes, each carrying 5 marks, set out of the entire syllabus would be compulsory. Additionally, the students are required to attempt one question from each Unit carrying 10 marks.

Unit-I

Introductory Econometrics: Introduction to Nature, Role and Significance of Econometrics, Notes of Caution while Developing and Using Econometric Models.
Classical Econometrics: Bi-variate and Multivariate Regression Analysis Using STATA/Eviews/SPSS/Gretl/R; interpretation of coefficients, test of hypotheses and restrictions, goodness of fit statistics. Testing the Existence and Solving the Problems of Non-normality, Heteroscedasticity, Autocorrelation and Multicollinearity using STATA/Eviews/SPSS/Gretl/R.

Unit – II

Econometrics of Categorical Variables: Role of Categorical Variables in Econometric Modeling. Creating Categorical/Polychotomous Variables in STATA/Eviews/SPSS/Gretl/R. Estimating and interpreting the models with categorical variables as independent variables in STATA/Eviews/SPSS/Gretl/R.

Categorical Dependent Variable Models: Models with Dichotomous Response; Logit, Probit and Tobit estimation using STATA/Eviews/SPSS/Gretl/R. Models with Polychotomous Response; Multinomial Logit/Probit and Ordered Response Models Estimation using STATA/Eviews/SPSS/Gretl/R. Interpretation of Coefficients and Marginal Effects. Test of Significance and Goodness of Fit Statistics for Categorical Response Variable Models in STATA/Eviews/SPSS/ Gretl/R.

Unit – III

Simultaneous Equation Models: Estimating Simultaneous Equation Models using STATA/Eviews/SPSS/Gretl/R. Basic Discussion on Identification Problem; Rank and Order Conditions for Identification. Estimating the Simultaneous equations models using Indirect Least Squares (ILS), Two Stage Least Squares (2SLS), Full Information Maximum Likelihood (FIML) and Seemingly Unrelated Regression (SUR) methods in STATA/Eviews/SPSS/Gretl/R.

Time Series Analysis*:* Using Distributed Lag, Autoregressive, and Moving Average terms in STATA/Eviews/SPSS/Gretl/R. Testing and Interpreting the Presence of Unit-root, Seasonality and Structural Breaks in Time Series Variables using STATA/Eviews/SPSS/Gretl/R. Estimating and Interpreting Co-integration, VAR/VECM and Impulse Response models/coefficients using STATA/Eviews/SPSS/Gretl/R.

Unit – IV

Panel Data Analysis*:* Testing Significance of Cross-section and Time-specific Effects in STATA/Eviews/SPSS/Gretl/R. Estimation and Interpretation of the Coefficients of Fixed and Random Effect Models in STATA/Eviews/SPSS/Gretl/R. Hausman Specification Test and Model Selection criteria in STATA/Eviews/SPSS/Gretl/R. Random Coefficient models estimation using STATA/Eviews/SPSS/Gretl/R. An Introduction to Panel Data Unit-root and Co-integration testing using STATA/Eviews/SPSS/Gretl/R.

Suggested Readings :-

1. Damodar Gujarati (2003). *Basic Econometric.* (4th ed.). McGraw Hill, New York.
2. Jack Johnston and John Dinardo (1997). *Econometric Methods.* (4th ed.), McGraw Hill, New York.
3. Hill, Carter.R., Griffiths, W.E. and Lim.G.C. *Principles of Econometrics* (4th Edition), John Wiley and Sons.
4. Adkins, Lee.C. and Hill, Carter.*R Using STATA for Principles of Econometrics ,* (4th ed.), John Wiley and Sons.
5. Griffiths, W.E., and Hill, Carter.R. *Using Eviews for Principles of Econometrics*. (4th ed.), John Wiley and Sons.
6. Griffiths, W.E., and Hill, Carter.R.. *Using Gretl for Principles of Econometrics* (4th ed.), John Wiley and Sons. Available at www.learn**econometrics**.com/**gretl/using_gretr**_POE4.pdf
7. Studenmund, A.H. *Using Econometrics: A Practical Guide* (7th ed.), Pearson IndiaEducation Services Pvt. Ltd.

PAPER – III: OPTION (ii): ECONOMIC THEORY

4 Credits
100 marks
(50+50 internal assessment)
Teaching Hour: 40
Times: 3 Hours

Objectives: The aim of this paper is to make the students familiar with some of the recent developments in micro and macro economics which have wide range of applicability to solve and explain phenomena; some of which can provide new insights into the empirical estimation of economic relationship.

Pedagogy: Class Room Teaching, Term Paper/Assignment, Presentations, Book Reviews & Seminars.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The 1st question consisting of 2 short notes, each carrying 5 marks, set out of the entire syllabus would be compulsory. Additionally, the students are required to attempt one question from each Unit carrying 10 marks.

UNIT – I

Missing markets: some applied micro economic aspects

Externalities: Bilateral and Multilateral externalities, Inefficiency of resource allocation in the presence of externalities; Private Cost versus Social Cost.

Solutions to the problem of externalities: Government Responses; Regulation, Corrective Taxes, The Compensation principle, Creating a market. Private Responses; Allocating Property Rights, Bargaining and Coarse Theorem.
Non convexities and the Theory of Externalities

UNIT – II

Public Goods: Efficient provision of discrete and continuous public goods. Private provision of discrete and continuous public goods. Voting for Public goods. Lindahl allocations.

Asymmetric Information:
Issues of Hidden characteristics/actions and Moral Hazard; The Principal Agent problem. Moral Hazard in Insurance and Product markets.
Adverse Selection and the problem of lemons; the efficiency effects of adverse selection Market and Government responses to Asymmetric Information; Market Signaling

UNIT – III

Monetarist Keynesian Debate of Macroeconomic Theory and Policy.
The Structure, Policy implications and Assessment of the New Classical school,

The core propositions, features, policy implications and Assessment of New Keynesian School.
Essential features of the Austrian school

UNIT – IV

Exploring Money Multiplier, Demand for Money and Philips Curve in Indian context. Macro Economic theory for India

The Global Economic Crisis and Macro Economic theory India and the Global Economic Crisis.

Suggested Readings :-

1. Ben Fine, Costas Lapvitsas and Jonathan Pincus eds. (2001) Development Policy in the Twenty First Century Beyond the Post- Washington Consensus, Routledge, Oson.
2. Bhaduri, A (1993) Unconventional Economics Essays, Oxford University Press, New Delhi.
3. Dornbusch, R, S. Fischer, R Starz (2004) Macroeconomics, (9th ed.) Tata McGraw Hills Publishing Co. Ltd., New Delhi.
4. Eric J. Pentecost (2000) Macroeconomics: An open economy approach, Macmillan Press Ltd., London.
5. Gordon Robert J. (2012) Macro Economics, Pearson Education.
6. Gupta, G S (2004) Macroeconomics: Theory an Applications (2nd Edition), Tata McGraw Hills Publishing Co. Ltd., New Delhi.
7. Gupta, P.K. and Heera, D.S. Operations Research, Sultan Chand.
8. Griffiths, W.E., and Hill, Carter. R :Using Gretl for Principles of Econometrics [4thEdition], John Wiley and Sons.
9. Jha Raghbendra (2008) Contemporary Macroeconomic Theory and Policy, New Age International Publication.
10. Katz, M L and Harvey S Rosen Microeconomics (2nd edition). (1999)
11. Krishna, K.L. (1999) Econometric Application in India, Oxford University Press, New Delhi.
12. Kumbhakar, Subal C. (1994) Stochastic Frontier Analysis, Cambridge, University Press.
13. Mas-Colell, M.D. Whinston & J.R. Green (1995) Microeconomic Theory, Oxford University Press.
14. Patnaik, P (1997) Macroeconomics, Oxford University Press, New Delhi.
15. Pindyck, R.S.Rubinfeld, D.L & P.L. Mehta (2005) Microeconomics (5th Edition),Pearson Education Delhi.
16. Rakshit, M (1989) Studies in the Macroeconomics of Developing Countries, Oxford University Press, New Delhi.
17. Ramanathan, R. An Introduction to Data Envelopment Analysis: A Tool for Performance Measurement, Sage Publications.
18. Rodrik, Dani (2007) One Economics, Many Recipes: Gobalization, Institutions and Economic Growth, Princeton University Press.
19. Rostow, W W (1991) The Stages of Economic Growth: Communist Manifesto, Cambridge University Press.
20. Solow, Robert M (2000) Growth Theory: An Exposition, Second Edition, Oxford University Press, Inc.
21. Snowdon Brain and Howard R. (2005) Modern Macroeconomics, Vane Edward Elgar
22. Thirlwal, A (2005) Growth and Development: With Special Reference to Developing Economies, Palgrave Macmillan.

PAPER- III: OPTION (iii): ECONOMIC DEVELOPMENT

4 Credits
100 marks
(80+20 internal assessment)
Teaching Hour : 40
Time : 3 Hours

Objectives: The global developments of the past about two decades have thrown open enormous opportunities in diverse forms. They have created several daunting challenges as well. Research is perceived to play very important role in enabling communities and policy makers to cope up with the emerging challenges successfully. Against this backdrop, the basic objective of this course is to make the prospective researchers familiar with the development issues currently facing the global community. It aims to equip them with the latest developments in the area so that they get ignited to undertake socially meaningful research, particularly in the emerging scenario.

Pedagogy: Class Room Teaching, Term Paper/Assignment, Presentations, Book Reviews & Seminars.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The 1st question consisting of 2 short notes, each carrying 5 marks, set out of the entire syllabus would be compulsory. Additionally, the students are required to attempt one question from each Unit carrying 10 marks.

UNIT – I

Development Issues of our Time I: Looking beyond the Neo-classical Growth Theory; Role of Human Capital (Health, Education), Innovations and Evolving Technologies, Environmental Issues.

UNIT – II

Development Issues of our Time II: Institutions and Development; Markets, Networks, Social Capital; Finance and Development; Political Economy and Policies of Development; Inequality and Development.

UNIT – III

Development Policy in the Twenty-First Century-I: Creating Roadmaps towards Quest for Happiness through Inclusive Growth; Creating more Gainful Employment and Earning Opportunities, Devising Mechanisms for Empowerment.

UNIT – IV

Development Policy in the Twenty-First Century-II: Alternate Policy Paradigms: Economic Nationalism in a Globalizing World; Cooperation versus Competition and Development.

Suggested Readings:-

1. A B Atkinson and A Brandolini (2001), "Promise and Pitfalls in the use of " Secondary" Data Sets: Income Inequality in OECD Countries, " Journal Of Economics Literature.
2. Ades, Alberto and Edward Glaeser,'' Evidence on growth, increasing returns, and the extend of the market ,"Quarterly Journal of Economics, Volume CXIV (3), August 1999,pp.1025-1046.

Aghion and Howitt, Chapters 2,3,12.

3. Alberto Alesina and Beatrice Weder, Do Corrupt Governments Receive Less Foreign Aid, Revised: May,2000,http:/post.economics.havard.edu/faculty/alesina/pdf-papers/aidjune 10.pdf Alberto Alesina and Dani Rodrik (1994),"Distributive politics and economic growth ," Quarterly Journal of Economics,108: 465-90.
4. Alberto Alesina and Roberto Perotti."The Policies of Growth : A Survey of the Recent Literature,"World Bank Economics Review, September 1994 8:351-72
5. Allan Drazen, Ploitical Economy In Macroeconomics, Princeton University Press, 2000, Chapter 10 and 11.
6. Barro and Sala- Martin, Chapter 3-5.
7. Barro and Sala- i-Martin, Economic Growth, see especially Chapters 1 & 2.
8. Bils, Mark and P. Klenow "Does Schooling Cause Growth? 'American Economic Review, December 2000, 90(5), pp.1160-1183.
9. Burnside, Craig and David Dollar," Aid Policies, and Growth,"Amerrican Economic Review 90(4) (September2000): pp.847-68.
10. Chang Tai Hsieh and Peter Klenow, Relative Prices and Relative Prosperity, mimeo Stanford and Berkeley, August 2004.
11. Chang Tai Hsieh, What Explains the Industrial Revolution in East Asia? Evidence from the factor Markets. (American Economic Review, june 2002). http:/www.princeton.edu/-chsieh/aea3pdf
12. Conley, Timothy and Chris Udry.2002.Learning about a New Technology: Pineapple in Ghana. Manuscript:Yale.http://www.econ.yale.edu/~udry/pdf/learn.pdf
13. Costas Azariadis and John Stachurski, Poverty Traps?,Prepared for the Hnadbook of Economic Growth, 204,http:/elsa.berkley.edu/~chad/azstach.pdf
14. Dani Rodrik, Growth Strategies, a paper for the Handbook of Economic Growth, April 2003.
15. Daron Acemoglu, Simon Johnson, James Robinson, Institutions as the Fundamental Cause of Long-Run Growth, April 29,2004, forthcoming Handbook of Economic Growth, http://else.berkeley.edu/~chad/handbook9sj.pdf
16. Daron Acemoglu, Simon Johnson, james Robinson, Reversal of Fortune: Geography and Institutions in the making of the modern World Income Distribution, Quarterly journal of Economics, November 2002, volume117, pp.1231-1294.
17. Daron Acemoglu, Simon Johnson, James Robinson, The Colonial Origins of Comparative Development: An Empirical investigation, American Economic Review, December, 2001, volume91, pp.1369-1401.
18. David Albouy, The Colonial Origins of Comparative Development: A Reinvestigation of Data, Department of Economics, University of California-Berkeley, july 2004.
19. Deepa Narayan, and Lant Pritchett, Cents and Sociability: Household of Income and Social Capital in Rural Tanzania, Economic Development and Cultural Change,1999, Vol.47(4).
20. Edward L.Glaeser, BruceI.Sacerdote and jose A Scheinkman The Social Multiplier (June 2002) European Economic Review, Forthcoming.
21. Edward L.Glaeser, David Laibson, and Bruce Sacerdote the Economic Approach to Social Capital, Economic Journal,112 (2002): 437-458.
22. Edward L.Glaeser, Rafael La Porta, Florecio Lopez-de-Silanes, Anderi Shleifer, DO INSTITUTIONS CAUSE GROWTH? NBER Working Paper 10568, june 2004, http://papers.nber.org/papers/w10568.pdf
23. Esther Duflo and Abhijit Banerjee Inequlitiy and Growth: What Can the Date Say? June 2003, Revised, http://econ-www.mit.edu/faculty/download_pdf.php?id=445
24. Forbes, Kirsten J.(2000)" A Reassessment of the relationship between inequality and growth"

American Economic Review,90(4) ,869-887.
25. Francesco Caselli " The missing input: Accounting for Cross-Country Income Differences", forthcoming chapter in the Handbook of Economic Growth November 2003, http://post.economics.havard.edu/faculty/caselli/papers/handbook.pdf
26. Grief, Avner (1993) "Contract Enforceability and Economic Institutions in Early Trade: The Maghribi Traders," American Economic Review,83 (3),pp.525-548.
27. Jacob Svensson, " Foreign Aid and Rent –Seeking", Journal of International Economics,2000,vol.51 (2):437-461.
28. Jacob Svensson, Why Conditional Aid Doesn't Work and What Can Be Done About It? Journal of Deveopment of Economics,2003, Vol.70 (20:381-402).
29. James Rauch, "Business and Social Networks in International Trade," "Journal of Economic Literture 39 (December 2001): 1177-1203.
30. Jess Benhabib and Mark M. Spiegel, Human capital and Technology Diffusion, December 9,2002,http://www.econ.nyu.edu/user/benhabib/growthhandbook10.pdf
31. Jones, (Chad.1995a) " R&D-Based Models of Economic Growth", journal of Political Economy, August,103,pp.759-84.
32. Jones, Charles.(1995) " Time Series tests of Endogenous Growth models"Quartely journal of Economics, May, 105(2), 495-526.
33. Klenow, Peter and Andres Rodriquez-Clare (1997b). "The Neoclassical Revival in Growth Economics: Has it Gone Too Far?" NBER Macroeconomics Annual 1997, Volume 12, 73-103.
34. Krueger, Alan B. And Mikael Lindahl " Education For Growth" Journal of Economic Litreature, 2001.
35. Lant prichett, Diveergence, Big Time. Journal of Economic Perspectives, 1997.
36. Mankiw, N.Gregory (1995). "The Growth of Nations", Brookings Papers on Economic Activity 1, pp.275-326.
37. Marcel Fafchamps, Market Institutions in sub-Saharan Africa, MIT Press, 2004 (chapter 1,2,22, and possibly others)
38. Marcel Fafchamps, Networks, Communities and markets in SUB-Saharan Africa: Implications of firm Growth and investment, journal of African Economics, Volume 10, supplement 2 : September 2001,pp.109-142.
39. Oded Galor; Joseph Zeira, Income Distribution and Macroeconomics, The Review of Econmics Studies, Vol.60, no.1.(Jan.,1993),pp.35-52.
40. Peter Boone, " Politics and the Effectiveness of Foreign Aid." European Economic Review, 1996,40(2),pp.289-329.
41. Francesco Caselli " The missing input: Accounting for Cross-Country Income Differences", forthcoming chapter in the Handbook of Economic Growth November 2003.http://post.economics.havard.edu/faculty/caselli/papers/handbook.pdf
42. Grief, Avner (1993) "Contract Enforceability and Economic Institutions in Early Trade: The Maghribi Traders," American Economic Review,83 (3),pp.525-548.
43. Jacob Svensson, " Foreign Aid and Rent –Seeking", Journal of International Economics,2000,vol.51 (2):437-461.
44. Jacob Svensson, Why Conditional Aid Doesn't Work and What Can Be Done About It? Journal of Deveopment of Economics,2003, Vol.70 (20:381-402).
45. James Rauch, "Business and Social Networks in International Trade," "Journal of Economic Literture 39 (December 2001): 1177-1203.
46. Jess Benhabib and Mark M. Spiegel, Human capital and Technology Diffusion, December 9,2002,http://www.econ.nyu.edu/user/benhabib/growthhandbook10.pdf
47. Jones, (Chad.1995a) " R&D-Based Models of Economic Growth", journal of Political Economy, August,103,pp.759-84.
48. Jones, Charles.(1995) " Time Series tests of Endogenous Growth models"Quartely journal of Economics, May, 105(2), 495-526.
49. Klenow, Peter and Andres Rodriquez-Clare (1997b). "The Neoclassical Revival in Growth Economics: Has it Gone Too Far?" NBER Macroeconomics Annual 1997, Volume 12, 73-103.
50. Krueger, Alan B. And Mikael Lindahl " Education For Growth" Journal of Economic Litreature,

2001.
51. Lant prichett, Diveergence, Big Time. Journal of Economic Perspectives, 1997.
52. Mankiw, N.Gregory (1995). " The Growth of Nations", Brookings Papers on Economic Activity 1,pp.275-326.
53. Marcel Fafchamps, Market Institutions in sub-Saharan Africa, MIT Press, 2004 (chapter 1,2,22, and possibly others)
54. Marcel Fafchamps, Networks, Communities and markets in SUB-Saharan Africa: Implications of firm Growth and investment, journal of African Economics, Volume 10, supplement 2 : September 2001,pp.109-142.
55. Oded Galor; Joseph Zeira, Income Distribution and Macroeconomics, The Review of Econmics Studies, Vol.60, no.1.(Jan.,1993),pp.35-52.
56. Peter Boone, " Politics and the Effectiveness of Foreign Aid." European Economic Review, 1996,40(2),pp.289-329.
57. Pritchett, Lant (2001). " Where Has All The Education Goner?" World Bank Economic Review.
58. Pritchett, Lantt. " Does Learining to add up? The return to Schooling in aggregate data," draft for Handbook of Education Economics, September 2,2004.
59. Romer, Paul (1995). Comment on N. Gregory mankiw, "The Growth of the Nations" Brooking Papers on Economic Activity 1,pp.313-320.
60. Ronald Benabou, Inequality, Technolohgy, and the social Contract, December 2003, chapter for Handbook of Economic Growth,http://www.princcton.edu/%Erebenabou/Handbook1.pdf
61. Ross Levine, " Finance and Growth: Theory and Evidence", Chapter for Handbook of Economic Growth, September 2004, available at http://legancy.csom.umn.edu/WWWPages/FACULTY/RLevine/Index.html
62. Stanly L. Engerman and Kenneth L. Sokoloff, Factor Endowments, Inequality, and paths of Development among New World Economies, Economia, Fall2002
63. Thorsten Beck and Ross Levine, Legal Institutions and Financial Development, July 29,2003, available at http://legancy.csom.umn.edu/WWW Pages/Faculty/R
64. Valerie Bockstette, Arrendam Chanda, and Louis Putterman,2002, States and Markets: The Advantage of an Early Start, journal of Economic Growth,7,347-369.
65. W. Easterly and R.Levine, " Africa's Growth Tragedy: Policies and Ethnic Divisions," November 1997, Quaterly Journal of Economics, CXII(4),1203-1250.
66. W. Easterly, " National Policies and Economic Growth: A Reappraisal" Development Research Institue (DRI) Working Paper No. 1, New York University, paper for the Handbook of Economic Growth, March,2003.
67. William Easterly, Ross Levine, and David Roodman " New Data, New Doubts: A Comment on Burnside and Dollar's " Aid, Policies, and Growth," (2000). American Economic Review, 2003, forthcoming.

Additional Readings

1. Amartya Sen (1999), " Employment, Technology and Development". Oxford University Press.
2. Ben Fine, Costas Lapvitsas and Jonatham Pincus eds.(2001), Development Policy in the Twenty-First Century Beyond The Post-Washington Consensus, Routledge, Oson Banka Milanovic (2007), Worlds Apart: Measuring International and Global Inequality, Princeton University Press.
3. Dani Rodrik (2007), One Economics, many Reciepes: Globalization, Institutions and Economic Growth, Princeton University Press.

4. Igancy Sachs (2000), understanding Development, Oxford University Press.
5. Jeffriy D. Sachs(2005), The End of the Poverty: Economic Possibities for our Time, Penguin Press.
6. Manus I. Midlarsky ed.(1997) , Inequality, Democracy, and Economic Development, Cambridge University Press.
7. Narcis Serra, Joseph E. Stiblitz Eds.(2008), The Wsigton Consensus Reconsidered: Towards a New Global Govenance (Initiative for policy Dialouge), Oxford.
8. Paul Patrick Streeten (1997), Thinking about Deveopment, Cambridge University Press. Sen, A.K.(1984), Resources, Values and Development, Basil Blackwell, Oxford.
9. Sen, A.K.(1999), Development As Freedom, Knopf, Harvard. Toye, J.(1987), Dilemmas of Development, Balckwell, Oxford.
10. World Bank (2005), World Development Report: Equity and Development, World Bank, The U.S.
11. World Bank(2001), Finance for Growth: Policy Choices in a Volatile World, Oxford University Press Inc, U.S.A.
12. W.W. Rostow (1991), The Stages of Economic Growth: A Non-Communist Manifesto, Cambridge University Press.

SEMESTER-II

PAPER IV & V: OPTION (i) PUBLIC POLICY AND GOVERNANCE

4 Credits
100 marks
(50+50 internal assessment)
Teaching Hour: 40
Time: 3 Hours

Objectives : - Policy Analysis has increasingly evolved into a structured set of steps that help policy makers, decision makers in the government, and society at large, to explore ways to improve the provision of public goods and services. The study of Public Policy process with a special focus on economic policy and governance is imperative to build the capacity to conduct research in economic issues of contemporary relevance.

Pedagogy: The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The 1st question consisting of 2 short notes, each carrying 5 marks, set out of the entire syllabus would be compulsory. Additionally, the students are required to attempt one question from each Unit carrying 10 marks.

Unit-I

Role of the State in Policy Making: Market Failure and Need for Government Intervention. Government Failure. Alternatives to Market and Government Failure. Meaning and Role of Centre of Government.

Unit-II

Policy Process: Need for Public Policy. Policy making process. Models of Public Policy - Institutional, Process, Incremental, Group, Elite, Public Choice and Game Theory model. Attributes of a good policy making process.

Unit-III

Governance: Concept and Need for Good Governance. Dynamic Governance. Transparency and Governance. Concept of Fragile State – Failed Governance.

Unit-IV

Economic Policies: Economic theories as policy guides. Economic Governance. Governance Issues in India's Economic Reforms . Governance and Policy Implementation in India.

Suggested Readings:-

1. R. Dye, Thomas. (2002). *Understanding Public Policy*, Pearson Education, New Delhi.
2. Nagel, Stuart S., (1984). *Public Policy – Goals, Means and Methods*, St. Martin's Press, N.Y.
3. Nagel, Stuart S., (2000). *India's Development and Public Policy*, Ashgate Publication Co.
4. Hill, M and Hupe, P (2002).*Implementing Public Policy,* Sage, London.
5. World Bank (2017). *World Development Report*: 'Governance and the Law' Washington D.C.
6. Government of India. Twelfth Plan Document Several Articles and Working Papers.

PAPER IV&V OPTION (ii) – ECONOMIC METHODOLOGY

4 Credits
100 marks
(50+50 internal assessment)
Teaching Hour: 40
Time: 3 Hours

Objectives: - The economic crisis that has affected the world economy since 2008 has unfolded deep flaws in the content and the teaching methods of economics. Courses such as economic methodology and history of economic thought have slowly been removed from the economics curriculum around the world. This new module provides students the opportunityto participate in the revival of interest in studying economic methodology and history of economic ideas. The study of the evolution of economic ideas and of the economics profession remains relevant to understand how theory and methods have developed historically in economics. The course will bring the necessary elements of plurality and interdisciplinary that economics research students need in order to understand the depth of economic ideas and policy stances but also that there is no absolute economic truth and thatwe need to contest existent and accepted economic wisdom.

Pedagogy: Class Room Teaching, Term Paper/Assignment, Book Reviews and Seminar.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The 1st question consisting of 2 short notes, each carrying 5 marks, set out of the entire syllabus

would be compulsory. Additionally, the students are required to attempt one question from each Unit carrying 10 marks.

Unit-I

Evolution of economics as social science; Economic as a scientific discipline, Methods of economic analysis: Relativist, absolutist, deductive and inductive.
Four thesis on the state of modern economics; Sociology of economics.

Unit-II

Science and ideology in economics: Essays of Schumpeter and Solow.
Robbins' essay and positivism in economic methodology and its philosophical andideological critique.
Neoclassical Economics and its critique; Rationality in economic theory and its critics.

Unit-III

Friedman's methodological instrumentalism: Philosophical evaluation and rejection of instrumentalism, Ideological and methodological critique of Friedman's position.
Keynes on economic modes and econometrics: Debate with Tinbergen

Unit-IV

Marx's position on ideology and method in economics; Dialectical and historical materialismas method of inquiry; Mode of production and superstructure model.
Methodology of heterodox economic tradition.

Suggested Readings

1. Mark Blaug (1997), *Methodology of Economics: How Economists Explain,* Cambridge University Press.
2. Daniel M. Hausman Eds. (1984), *The Philosophy of Economics: An Anthology*, Cambridge University Press.
3. Bruce J. Caldwell (2003), *Beyond Positivism: Economic Methodology in the 20th Century*, Routledge Publication.
4. Milton Friedman (1953*), Essays in Positive Economics*, University of Chicago Press. Joseph Schumpeter (1976), *History of Economic analysis*, Oxford University Press. Tony Lawson (2003), *Reorienting Economics*, Routledge Books, London.
5. Tony Lawson (2013), What is this 'school' called neoclassical economics? *Cambridge Journal of Economics*, Vol. 37, No. 5, pp. 947-983.
6. Frederic S. Lee and Bruce Cronin Eds. (2016), *Handbook of Research Methods andApplications in Heterodox Economics,* Edward Elgar Publishing Ltd.
7. John M. Keynes (1939), Professor Tinbergen's' Method, *The Economic Journal*, Vol. 49, No.195, pp. 558-577.
8. Amaryta K. Sen (1977), Rational Fools: A Critique of the Behavioral Foundations of Economic Theory, *Philosophy & Public Affairs*, Vol. 6, No. 4, pp. 317-344.
9. Richard D. Wolff and Stephen A. Resnick (2012*), Contending Economic Theories:Neoclassical, Keynesian and Marxian*, The MIT Press.

PAPER IV: OPTION (iii): INDUSTRIAL ECONOMICS

4 Credits
100 marks
(50+50 internal assessment)
Teaching Hour: 40
Time: 3 Hours

Objectives: In the 21 century, efficiency and competitiveness in all spheres of economic life and more 500 industrial sectors are at the core of success. The present syllabus is designed for the students to capture the contemporary issues in industrial development bothin theory and in the Indian context.

Pedagogy : The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The 1st question consisting of 2 short notes, each carrying 5 marks, set out of the entire syllabus would be compulsory. Additionally, the students are required to attempt one question from each Unit carrying 10 marks.

Unit – I

Proprietorship versus Corporations; Structure Conduct Performance Paradigm of Brains; Collusion versus Efficiency Hypothesis; Dominant firm Behavior: Limit Price versus Strategic Conduct. Rivalry and Strategic Conduct.

Unit- II

Porter's Five Competitive Forces: Bargaining Power of Customers, The Bargaining power of Suppliers, The Threat of New Entrants and the Threat of Substitute Products - combined with other variables to Influence a Fifth force, The Level of Competition in an Industry.

Unit – III

Technological Progress: Concepts, Optimal Technological Change and optimal timing of Innovation. Importance of small versus large firms. Learning by doing and Size of the Firms. Logic of Patents.

Unit – IV

Public Policy for business combinations with special reference to India: Takeovers code of SEBI.

Empirical studies: Market Structure, Market Power and Profitability; Mergers and Takeovers in India. Profitability Issues, R & D behavior of firms. Market Structure and R & D Behavior of firms.

Suggested Readings:-

1. Martin, Stephen (2001). *Advanced Industrial Economics*, Blackwell, (2nd ed.).
2. Scherer, F M & Ross David (1990). *Industrial Market Structure and Economic Performance,*

(3 ed.). Boston Houghton Mifflin Company.
3. Hay, D A & Morris, D J (1979). *Industrial Economics and Organization,* Oxford Univ. Press,New York.
4. Shepherd, William, G, J.M. Shepherd (2003). *The Economics of Industrial Organization Analysis, Market Policies*. (5th ed.).Prentice Hall International.
5. Kreps. D.M, (2003). Microeconomics for Managers (1st ed.). w.w. Norton & Company, New York London.
6. Schmalensee, R and Robert Willig (1989). *Handbook of Industrial Organization* (Vol. I& II), Elsevier, North Holland.
7. P L Beena (September 27, 2008) *Economic & Political Weekly*; Trends and Perspectives on Corporative Mergers in Contemporary India.
8. Pramod Mantravadi, A Vidyadhar Reddy (September 29, 2007) *Economic & Political Weekly*; Relative Size in Mergers and Operating Performance: Indian Experience.
9. Pramod Mantravadi, A Vidyadhar Reddy (September 27, 2008) *Economic & Political Weekly*; Type of Merger and Impact on Operating Performance: The Indian Experience.
10. Nagesh Kumar (August 5, 2000). *Economic & Political weekly*; Mergers and Acquisitions by MNEs: Patterns and Implications.

PAPER IV& V: OPTION (iv) - MONEY, BANKING AND FINANCE

4 Credits
100 marks
(50+50 internal assessment)
Teaching Hour: 40
Time: 3 Hours

Objectives : This course aims to provide the doctoral students exposure to the recentdevelopments in money, banking and financial issues in Indian economy.

Pedagogy: Class Room Teaching, Term Paper/Assignment, Presentations, Book Reviews & Seminars.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The 1st question consisting of 2 short notes, each carrying 5 marks, set out of the entire syllabus would be compulsory. Additionally, the students are required to attempt one question from each Unit carrying 10 marks.

Unit – I

Money Inflation and Growth.
Reserve Bank of India and Money Supply Growth Rule.Estimating Money Multiplier for India.

Unit – II

Empirical Demand Function for money for India.
The Banking Sector in India: Emerging issues and challenges.
Recent Developments in the Theory and Role of Central Banks with special reference tothe issues of Role, Autonomy and Monetary Policy of Reserve Bank of India.

Unit – III

Recent Developments in the Theory of Financial Markets and Asset Pricing.Current state of the Indian Capital Market.

Unit – IV

Financial Liberalization in India: An Assessment.Financial Sector Regulation in India.

Suggested Readings :-

1. Benjamin M. Friedman and Frank H. Haun (2000) (ed.). *Handbook of Monetary Economics*; Volume I and Volume II.
2. Peter Bofinger (2001). *Monetary Policy: Goals, Institutions*, Strategies and Instruments, Oxford University Press.
3. Rakesh Mohan (2009). *Monetary Policy in a Globalized Economy*: A Practioner's View, Oxford University Press.
4. *Economic and Political Weekly*: Various Issues.
5. *The Indian Economic Journal*: Recent Issues.
6. *Government of India: Economic Survey*; Latest Issue.
7. *Reserve Bank of India*: Report on Currency and Finance, Current Issues.

PAPER IV: OPTION (v) - INTERNATIONAL ECONOMICS

4 Credits
100 marks
(50+50 internal assessment)
Teaching Hour : 40
Time : 3 Hours

Objectives: The course intends to provide a deep understanding about the broad principles and theories, which tend to govern the free flow of trade in goods, services and capital at the global level. The contents of the paper spread over different modules, lay stress on the theory and nature of the subject which, in turn, will greatly help them to examine the impact of the trade theories and policies followed both at the national and international levels also their welfare implications at macro level and the distribution of gains from trade.
The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students writing assignments, tutorials and seminars.

Pedagogy: Class Room Teaching, Term Paper/Assignment, Presentations, Book Reviews & Seminars

.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The 1st question consisting of 2 short notes, each carrying 5 marks, set out of the entire syllabus would be compulsory. Additionally, the students are required to attempt one question from each Unit carrying 10 marks.

Unit-I

Trade Based on Competitive Advantage: Imperfect Competition and Intra Industry Trade.
Economics of Services Trade.

Unit-II

Global Financial Institutions – Role, Functions and Working of International Financial Corporation, World Bank, International Development Association, International Financial Corporation and Asian Development Bank.

Unit-III

Financial Crises: Explanations, Types and Implications.
East Asian Crisis, 2008 US Recession and Eurozone Debt Crisis.

Unit-IV

Global Environmental Problems: Global Warming Ozone Depletion and Acid Rain (Consequences and Solutions or International Responses); Earth Summits 1992, 2002 and 2012.
Trade and Environment: Environmental Effects of International Trade;Environmental Concerns, the Global Commons and the Indian Economy.

Suggested Readings :

1. Claessens, Satijn, M.Ayhan Kose., Luc Laeven and Fabian Valencia. (2014). *Financial Crises: Causes, Consequences and Policy Responses,* IMF, Academic Foundations.
2. Frankel, Jeffrey. (2009). *Environmental effects of International Trade*, Faculty Research working paper Series. (John F. Kennedy School of Govt. – Harvard University).
3. Ghosh, Jayati. (2015). Economics: *India and the International Economy* (Vol.2), Oxford University Press: 440-469.
4. Hussen , Ahmed.(2004). *Principles of Environmental Economics,* Routledge, Taylore & Francis group, 2nd edition, London & New York.
5. Krugman, P.R. and Maurice Obstfeld. (2016). *International Economics: theory and Policy*, Eighth edition, Pearson education Inc.
6. Lukauskas, Arvind., Robert M. Stern., and Gianni Zanini. (2013). *Handbook of trade Policy for Development*. Oxford University Press.
7. Marrewi jk, Charles Van. (2017) *International trade*, Oxford University Press
8. Sharma, Shalendra D. (2014). *Global Financial Contagion: Building a Resilient World Economy after the Subprime Crisis.* Cambridge University Press.
9. Stevens, Candice (1993). The environmental effects of trade, *The world economy*; 16(4): 439-451.
10. www.worldbank.org
11. www.imf.org
12. www.adb.org
13. www.ifc.org
14. www.ida.worldbank.or

PAPER IV&V: OPTION (vi) AGRICULTURAL ECONOMICS

4 Credits
100 marks
(50+50 internal assessment)
Teaching Hour : 40
Time : 3 Hours

Objectives: The paper is designed to expose the students to the contemporary issues in agricultural development both in the theory and in the Indian context.

Pedagogy: The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The 1st question consisting of 2 short notes, each carrying 5 marks, set out of the entire syllabus would be compulsory. Additionally, the students are required to attempt one question from each Unit carrying 10 marks.

UNIT – I

Agriculture Transformation – Role of Agricultural Innovations: Generations of Innovations (induced innovations, Political Economy of Public Funded Innovations, and Policies and Institutions Managing Innovations]; Technology Adoptions [Adoption and Diffusion, Risk Management, Institutional Constrains to Innovations and Farm Policy]; Ruttan-Hayami Induced Innovation Hypothesis.

Agricultural Commodity Futures and Options: Nature and Principles of Future Markets; Participants in Future Markets; Trading in the Presence of Risk Aversions and Transaction Costs, Future Prices as Forecasts.

UNIT – II

Agro-processing: Specificities of Agro-processing, Developmental Role of Agro-processing Industries, Selection and Evolution Options; Food Processing Policy of India, 2005.

Agricultural Finance: Agency Relationship, Adverse Selection, Moral Hazards and Enforcement Problem; Segmented Rural Credit Markets; Resolving Information and Incentive Problems; Managing Borrowers' Credit Risks, Agribusiness and Trade Financing and Role of Social Capital.

Unit – III

Political Economy of Agricultural Policies: Protection Across Countries and Sectors, Influence of Political Institutions; Collective Actions of Lobbing Groups and Strategic Interactions between Lobbing Groups and Politicians; Use of Inefficient Instruments and Rent Seeking; and Explaining Public Investment in Agricultural Research.

Food Security and Food Assistance: Concept, Threat, Indicators and Mechanism to Food Security; Food Assistance Programmes [both Domestic and International], and Their Effectiveness.

UNIT – IV

Indian Agriculture- Opportunities and Challenges: Expectations, Performance, Problems and Prospects of Indian Agriculture in the Post-reform Period.

Investment in Farm People: Role of Human Capital Formation [Education and Health] in Modernization and Transition of Agrarian Economies.

References:

1. Bruce L. Gardner and Gordon C. Rausser (2002) *Handbook of Agricultural Economics, Vol-1A,1B, 2A, 2B and 3;* North-Holland, Amsterdam.
2. James G. Brown and Deloitte and Touche (1994) *Agro-industrial Investment and Operations*, Washington, The World Bank.
3. Government of India (2005) *Food Processing Policy, Ministry of Food Processing*, New Delhi.
4. T.W.Schultz (1981) *Investing in Peoples: The Economics of Population Quality*, University of California, Berkley.
5. Radhakrishna R, S. K. Rao, S. Mahendra Dev and K Subbarao (2006) *India in a Globalising World: Some Aspects of Macro economy, Agriculture and Poverty,* Academic foundation, New Delhi.
6. Government of India (2004) *State of the Indian Farmer: A Millennium Study, Academic Foundation and Ministry of Agriculture,* New Delhi, (Vol. 1-27).

****